

What Federated KANs Actually Buy: Learning-Rate Robustness at a Compute Cost for NetFlow-Based IoT Intrusion Detection

Phuc Hao Do, *Member, IEEE*, Van Long Nguyen, Tran Duc Le, and Truong Duy Dinh

Abstract—Federated Learning (FL) for IoT intrusion detection (IDS) is bottlenecked both by the bandwidth budget of edge devices and by the accuracy degradation that occurs when client traffic distributions are heterogeneous. Recent work has integrated Kolmogorov–Arnold Networks (KANs) into FL outside the IDS setting; we present FedKAN-IDS, the first systematic, parameter-matched empirical study of Federated KANs have been proposed for federated intrusion detection on the promise that, at a matched parameter budget, they are more accurate and more stable than multilayer perceptrons (MLPs) under client heterogeneity. We tested that promise on three standardised NetFlow-v2 intrusion detection datasets (NF-BoT-IoT-v2, NF-ToN-IoT-v2, NF-CSE-CIC-IDS2018-v2), each evaluated with $n = 10$ seeds under three heterogeneity regimes and both binary and multi-class detection. Our contribution is empirical rather than methodological, and we are careful to scope it, with $n = 10$ seeds per cell, and it does not survive contact with a controlled comparison. (i) On binary detection under extreme heterogeneity (Dir $\alpha = 0.1$) FedKAN-8 ($\sim 3.3k$ parameters) achieves a $+5$ to $+6$ percentage-point mean advantage over a The reported advantage is confounded with how much hyper-parameter tuning each architecture receives. Sweeping the learning rate over six values for every architecture, the $+5.5$ pp mean advantage we ourselves reported on NF-BoT-IoT-v2 falls to $+0.8$ pp ($p = 0.74$) once the parameter-matched FedAvg-MLP on the two most class-skewed datasets, with the BoT-IoT-v2 worst-seed gap reaching $+24$ pp and a $2.6\times$ variance reduction; doubling MLP capacity does not close the gap, supporting an architectural rather than capacity-driven reading MLP is given its own best learning rate, which is an order of magnitude away from the shared value. Across all four cells, the advantage ranges from -0.5 to $+6.2$ pp and is dataset dependent, not architectural. (ii) On multi-class detection, FedKAN-8 attains a seed-paired $+1.73$ pp advantage on the most class-diverse dataset (CSE-CIC-IDS2018-v2) with $p = 0.012$ and bootstrap 95% CI $[+0.73, +2.74]$ excluding zero, the most rigorous single positive result in our study, and consistent with the prediction that each additional class amplifies the cumulative regularising effect of the KAN spline-bias term. The variance-reduction claim does not generalise: the MLP-to-KAN seed-variance ratio is $2.5\times$ on one dataset but $0.71\times$, $0.33\times$ and $0.87\times$ on the others,

that is, reversed. (iii) We identify a reproducible failure mode: seed 17's Dirichlet partition drives FedKAN-8 below What does hold on every dataset is robustness to the learning rate itself. Over a decade-wide band, KAN-8 loses at most 0.65, 1.58 and 2.53 pp from a bad choice while the parameter-matched MLP on two of three datasets, mitigated by either larger hidden width (F-KAN-16 escapes the trap) or a richer spline grid ($G=10$ recovers ~ 5 percentage points). All observations are anchored in a convergence analysis showing the KAN B-spline approximation introduces an additive $\mathcal{O}(G^{-(k+1)})$ bias term independent of the heterogeneity factor Γ_α . Code, configurations, losses 3.53, 6.38 and 289 reproducible runs are publicly available 6.72 pp, a factor of 2.7 to 5.4. For federated deployment across gateways, where per-client tuning is unavailable, this is the property that matters. (iv) It is not free. At matched parameters on single-thread CPU, KAN inference costs $19.9\times$ more per sample and $37.9\times$ more per batch of 1000; parameter parity is not cost parity. We therefore report a trade-off rather than a win, and release a comparison protocol (per-architecture tuning, multi-round summaries, and sensitivity of conclusions to the summary statistic) together with 1,280 reproducible runs.

Index Terms—Federated Learning, Kolmogorov-Arnold Networks, IoT Security, Intrusion Detection, Non-IID, Robustness

I. INTRODUCTION

THE pervasive deployment of Internet-of-Things (IoT) devices across smart cities, industrial control, health-care, and autonomous transport has dramatically expanded the network attack surface. Distributed denial-of-service (DDoS) campaigns, botnets, reconnaissance scans, and other intrusions traverse the same edge gateways that carry legitimate IoT traffic, motivating the deployment of network intrusion detection systems (IDS) at the network edge [1].

Federated learning for IoT IDS. Centralised IDS pipelines [2], in which raw flow records are exfiltrated to a central server for model training, raise persistent privacy concerns and conflict with regulations such as GDPR and HIPAA. Federated Learning (FL) [3], [4] offers a privacy-preserving alternative: each IoT gateway trains a local model on its own traffic and only the model updates are aggregated. A growing literature applies FL to IoT IDS, with notable contributions including DeepFed [5], which combined CNNs with gated recurrent units, and a wave of subsequent work on heterogeneity-robust aggregation, communication compression [6]–[8], and security-aware client selection.

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The communication-robustness tension. Two pressures shape FL-IDS design at the edge. First, IoT links are bandwidth-constrained: a typical LoRaWAN device has a duty-cycled uplink budget on the order of kilobytes per hour. Second, client data is highly non-identical. The geographically distributed nature of IoT deployments means that one gateway may see almost only DDoS traffic while another sees primarily reconnaissance scans, a scenario commonly modelled by a Dirichlet partition with a small concentration parameter α [9]. Under such partitioning, vanilla FedAvg-MLP architectures are known to suffer slow convergence and, more pathologically, occasional catastrophic divergence in which a single unfortunate client-initialisation pair drags the global model far from the consensus optimum.

Kolmogorov-Arnold Networks (KANs). The recent introduction of KANs [10] replaces fixed neuron-wise activations with learnable edge-wise B-spline activations. Unlike a dense MLP weight matrix, where each scalar parameter modulates a global linear interaction, each spline coefficient in a KAN layer has *local* support on the grid by construction of the B-spline basis. This locality has been argued to provide better interpretability and parameter efficiency in physics-discovery and scientific-computing settings. In the IoT security domain specifically, KAN variants have been benchmarked for botnet attack classification in the centralised setting [11]; the question this paper asks is whether that behaviour survives federation under client heterogeneity.

KAN-FL: a recently busy research line. Within a year of [10], several works have integrated KANs into federated learning. F-KANs [12] ported the original KAN to a vanilla FedAvg framework on CIFAR-style benchmarks. [13] performed the first systematic non-IID evaluation of FedKAN, reporting modest gains over MLPs at matched parameter budgets on tabular data. [14] extended the comparison across multiple aggregation strategies (FedProx, SCAFFOLD), and [15] applied a Fed-KAN to traffic prediction. [16] contributed a medical-imaging benchmark and [17] an ECG-signal study. Most recently, [18] proposed CG-FKAN, which sparsifies the KAN grid coefficients to attack communication efficiency ~~[-?]~~ directly.

Positioning of this paper. The methodological observation “KAN-FL converges in fewer rounds and transmits fewer bytes than a parameter-heavy MLP” has therefore been published several times outside the IDS setting. We do *not* claim that observation as new. What is new in this paper is (i) a systematic empirical study of FedKAN *specifically* on the standardised NetFlow-v2 family of IDS datasets [19], (ii) a parameter-matched comparison protocol that resolves the unfair-baseline ambiguity present in much prior FedKAN work, and (iii) a refined narrative anchored in a convergence analysis: FedKAN’s advantage is not in mean accuracy but in worst-case robustness under extreme client heterogeneity, and this advantage cannot be replicated by simply scaling up an MLP baseline.

Headline empirical finding. The advantage attributed to federated KANs in the recent literature, and in the first version of this manuscript, is confounded with how much hyper-parameter tuning each architecture receives. On

NF-BoT-IoT-v2 binary detection under Dirichlet $\alpha=0.1$ (extreme non-IID) with $K=10$ clients and $T=50$ rounds, averaged over $n=10$ seeds: $\alpha=0.1$, the parameter-matched FedAvg-MLP-PM (3, 362 parameters) collapses to as low as 56.5% accuracy on its worst seed, while FedKAN-8 (3, 280 parameters) maintains $\geq 80.7\%$ on every seed, with a 2.6 $\times$ smaller cross-seed standard deviation. The seed-paired bootstrap 95% confidence interval for the mean accuracy difference is $[-0.50, +14.57]$ percentage points. Doubling MLP capacity to 3.4k parameters *worsens* the worst-seed accuracy from 60.4% (small MLP at 1.3k params) to 56.5%, ruling out a capacity-driven explanation. MLP’s best learning rate is 10^{-1} , an order of magnitude away from the 10^{-2} shared by all architectures in the original comparison. Giving it that learning rate raises it by 5.39 pp and collapses the KAN advantage from +5.49 to +0.76 pp ($p=0.74$, paired over ten seeds). The same sweep on three further cells puts the advantage between -0.49 and $+6.23$ pp with no consistent sign, so the effect is a property of the dataset, not of the architecture.

Theoretical anchoring. We provide a convergence analysis (Theorem 1) showing that the FedKAN error decomposes into a standard FedAvg term proportional to the data-heterogeneity factor Γ_α plus an additive spline-approximation bias of order $\mathcal{O}(G^{-(k+1)})$. Crucially, the spline-bias term is independent of Γ_α , behaving as an effective regulariser whose relative impact *decreases* as heterogeneity grows. Empirically, this manifests as the smaller cross-seed variance we report.

Contributions.

- A negative result with a cause. We show that the KAN-versus-MLP gap reported for federated intrusion detection is largely a tuning artefact, and we quantify it: on the cell that produced our own headline number, tuning the baseline removes 5.39 of the 5.49 pp gap. We ~~present the first systematic~~, also show the companion variance-reduction claim reverses sign on two of four cells.
- A property that does generalise. Across all three datasets, KAN is markedly less sensitive to the learning rate than the parameter-matched ~~empirical comparison of FedKAN against FedAvg-MLP on the standardised NetFlow-v2 IDS datasets [19], under both binary and 5-class detection and three data-heterogeneity regimes (IID, Dirichlet $\alpha \in \{1.0, 0.1\}$).~~ Our protocol explicitly addresses the parameter-matching gap identified in prior FedKAN work.
- We identify a robustness/tail-risk advantage of FedKAN that is ~~architectural rather than capacity-driven: parameter-matched MLP suffers worst-seed accuracy drops of ~ 24 percentage points relative to FedKAN-8 under extreme non-IID, and doubling MLP capacity does not close this gap.~~ The advantage is supported by a seed-paired bootstrap 95% CI excluding zero across $n=10$ seeds., losing at most 0.65/1.58/2.53 pp from a bad choice within a decade-wide band against 3.53/6.38/6.72 pp, a factor of 2.7 to 5.4.
- We ~~anchor the empirical finding in a convergence analysis introducing a spline-bias term $\mathcal{O}(G^{-(k+1)})$ that~~

is independent of the heterogeneity factor and acts as an effective regulariser (Theorem 1 and Corollary ??). The price of that property. On single-thread CPU at matched parameters, KAN inference costs $19.9\times$ more per sample and $37.9\times$ more per batch of 1000, and KAN transmits more bytes to reach a fixed accuracy target. Parameter parity is not cost parity, and a paper that matches parameters has not thereby matched cost.

- We honestly walk back the “~50% communication reduction” framing of prior FedKAN-IDS work, showing via Proposition 2 that under parameter parity FedKAN incurs roughly 14% more bytes per round than its MLP counterpart, and reframe the case for FedKAN around tail-risk reduction, the more deployment-relevant metric for IDS.
- All code, dataset preparation scripts, configurations, and 289 reproducible runs are released at [A comparison protocol and the runs behind it](#). Per-architecture learning-rate tuning, accuracy summarised over the final five rounds rather than the last round alone, and an explicit test of how far each conclusion moves when the summary statistic changes. All 1,280 runs, all configurations and the analysis scripts are released.

Organisation. Section II introduces the KAN ~~++~~ layer formalism and the federated IDS optimisation problem. Section III details the FedKAN-IDS framework, including server and client algorithms. Section IV presents the four formal results that anchor our empirical narrative. Section V reports experimental protocol and results across architectures, partitions, and modes. Section VI discusses threats to validity and outlines next steps.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Kolmogorov–Arnold Network Layer

Kolmogorov–Arnold Networks (KANs) [10] replace the fixed neuron-wise activations of an MLP with *learnable* edge-wise activations parameterised as B-splines. Formally:

Definition 1 (KAN Layer). *Given input $\mathbf{x} \in \mathbb{R}^{n_{in}}$, grid size $G \in \mathbb{N}$, and B-spline order $k \in \mathbb{N}$, a KAN layer is the map $\Phi : \mathbb{R}^{n_{in}} \rightarrow \mathbb{R}^{n_{out}}$*

$$\Phi(\mathbf{x})_q = \sum_{p=1}^{n_{in}} \phi_{q,p}(x_p), \quad q = 1, \dots, n_{out}, \quad (1)$$

with each edge function

$$\phi_{q,p}(x) = w_{q,p}^{(b)} \sigma(x) + w_{q,p}^{(s)} \sum_{i=1}^{G+k} c_{q,p,i} B_i(x), \quad (2)$$

where $\sigma(\cdot) = \text{SiLU}(\cdot)$ is the residual base activation, $\{B_i\}_{i=1}^{G+k}$ is a B-spline basis of order k on a uniform grid of G points, and $\Theta_{KAN} = \{w_{q,p}^{(b)}, w_{q,p}^{(s)}, c_{q,p,i}\}$ are learnable.

By contrast, a fully-connected MLP layer with the same (n_{in}, n_{out}) employs a dense weight matrix $W \in \mathbb{R}^{n_{out} \times n_{in}}$, a bias $\mathbf{b} \in \mathbb{R}^{n_{out}}$, and a fixed scalar nonlinearity, so $\Theta_{MLP} = \{W, \mathbf{b}\}$. The architectural distinction relevant to our analysis is that each scalar weight in W controls a global linear

interaction, whereas each spline coefficient $c_{q,p,i}$ has *local* support on the grid, by construction of the B-spline basis.

B. Communication-Constrained Federated IDS

Definition 2 (Communication-Constrained Federated IDS). *Consider K IoT clients. Client C_k holds a private dataset $\mathcal{D}_k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_k}$ with feature vectors $\mathbf{x}_i \in \mathbb{R}^d$ and labels $y_i \in \{0, 1, \dots, c-1\}$, drawn independently from a per-client distribution $\mathcal{P}_k$. Heterogeneity across clients is modelled by drawing the per-class proportions of $\mathcal{P}_k$ from a Dirichlet distribution with concentration $\alpha > 0$ [9]: small α yields highly skewed $\mathcal{P}_k$ (extreme non-IID), while $\alpha \rightarrow \infty$ recovers the IID setting.*

Given a model $f(\Theta; \cdot)$ and a per-client empirical loss

$$\mathcal{L}_k(\Theta) = \frac{1}{N_k} \sum_{i=1}^{N_k} \ell(f(\Theta; \mathbf{x}_i), y_i), \quad (3)$$

the federated IDS objective is

$$\min_{\Theta} F(\Theta) = \sum_{k=1}^K \frac{N_k}{N} \mathcal{L}_k(\Theta) \quad \text{s.t.} \quad \sum_{t=1}^T \mathcal{C}_t \leq \mathcal{B}, \quad (4)$$

where $N = \sum_k N_k$, $\mathcal{C}_t$ is the per-round bytes summed across participating clients, and $\mathcal{B}$ is the deployment bandwidth budget.

In a synchronous FedAvg [3] solver of (4), each round t samples a fraction $C \in (0, 1]$ of the clients, transmits the global parameters Θ_t to each, runs E local epochs of stochastic gradient descent, and aggregates the resulting local parameters $\{\Theta_{t+1}^k\}$ by data-weighted averaging.

C. Related Work

We position FedKAN-IDS at the intersection of three active research threads.

(i) **FL for IoT intrusion detection.** DeepFed [5] introduced a CNN-GRU architecture for federated industrial IDS, demonstrating that FL can match centralised detection rates while preserving data locality. Subsequent surveys [1] catalogued the design space of FL-IDS and identified two persistent bottlenecks: communication overhead on bandwidth-constrained gateways, and accuracy degradation under non-IID partitioning of network traffic. Robust FL methods [6] have addressed the latter by reducing the effective heterogeneity through compressed communication, but the underlying client architecture has remained, almost without exception, a dense MLP or CNN.

(ii) **Kolmogorov–Arnold Networks and their federation.** The KAN architecture of Liu *et al.* [10] replaces fixed neuron-wise activations with learnable B-spline edge activations, providing better interpretability and parameter efficiency on smooth scientific-computing benchmarks. Within a year of that proposal, several works ported KAN to federated learning. F-KANs [12] integrated KAN into vanilla FedAvg on CIFAR-style image benchmarks. Sasse and de Farias [13] performed the first systematic non-IID evaluation of FedKAN on tabular data, reporting modest accuracy gains over MLP at

matched parameter budgets and proposing the same parameter-matching protocol we adopt here. Ma *et al.* [14] extended the comparison across multiple aggregation strategies (FedProx, SCAFFOLD), and Zeydan *et al.* [15] applied a Fed-KAN to traffic prediction. Lee *et al.* [16] contributed a medical-imaging benchmark and Zeleke and Bochicchio [17] an ECG-signal study. Most recently, Yu *et al.* introduced CG-FKAN [18], which sparsifies KAN grid coefficients to attack communication efficiency directly.

The methodological observation underlying this body of work that FedKAN converges in fewer rounds and transmits fewer bytes than a parameter-heavy MLP has therefore been published several times outside the IDS setting. Our contribution is *not* to claim that observation as new. Rather, we contribute (a) the first systematic study of FedKAN on the standardised NetFlow-v2 family of IDS datasets [19], (b) a parameter-matched experimental protocol that resolves an unfair-baseline ambiguity present in prior FedKAN work, and (c) a refined empirical claim that FedKAN’s advantage is robustness/tail-risk under heterogeneity rather than mean accuracy anchored in a convergence analysis with an explicit spline-bias term.

(iii) Communication-efficient FL. Our framing also intersects with the broader literature on communication-efficient FL. FedPAQ [7] reduces uplink bandwidth via periodic averaging plus stochastic quantisation; Top- k SGD [8] sparsifies gradient updates to the largest k entries; and [6] combined sparsification with non-IID-aware aggregation. CG-FKAN [18] can be viewed as a KAN-specific instantiation of this line. Our protocol does *not* apply any compression; we report all results in raw float32. As Section V makes explicit, under parameter parity FedKAN is in fact slightly *more* expensive to transmit than a comparable MLP, so the value proposition of our work is not communication efficiency per se but the heterogeneity-robust accuracy floor that FedKAN provides.

III. THE FEDKAN-IDS FRAMEWORK

This section describes how Definitions 1 and 2 are instantiated in the FedKAN-IDS pipeline. Section III-A fixes the per-client model architecture and the global system flow (Fig. 1); Section III-B formalises the server orchestration (Algorithm 1); Section III-C details the client local optimisation (Algorithm 2).

A. System Architecture

The FedKAN-IDS pipeline alternates between four stages per communication round, depicted in Fig. 1: (1) the server initialises the global KAN with hyperparameters (n_h, G, k) ; (2) broadcasts the current parameter triple $\Theta_t = \{w_t^{(b)}, w_t^{(s)}, c_t\}$ to the participating clients; (3) each client performs local optimisation against its private NetFlow records; and (4) the server aggregates returned parameters by data-weighted averaging.

Each IoT client C_k instantiates a single-hidden-layer KAN

$$f(\Theta; \mathbf{x}) = \Phi^{(\text{out})}(\Phi^{(\text{hid})}(\mathbf{x})), \quad (5)$$

Algorithm 1 FedKAN-IDS Server Orchestration

Require: Number of clients K , sampling fraction C , rounds T , KAN hyperparameters (n_h, G, k) .

Ensure: Trained global parameters Θ_T .

```

1: Initialise  $\Theta_0 = \{w_0^{(b)}, w_0^{(s)}, c_0\}$ 
2: for  $t = 0, 1, \dots, T - 1$  do
3:    $m \leftarrow \max(\lceil C \cdot K \rceil, 1)$ 
4:    $\mathcal{S}_t \leftarrow$  random subset of  $m$  clients from  $\{1, \dots, K\}$ 
5:   for each client  $k \in \mathcal{S}_t$  in parallel do
6:      $\Theta_{t+1}^k \leftarrow \text{CLIENTUPDATE}(k, \Theta_t)$ 
7:   end for
8:    $\Theta_{t+1} \leftarrow \sum_{k \in \mathcal{S}_t} \frac{N_k}{\sum_{j \in \mathcal{S}_t} N_j} \Theta_{t+1}^k$   $\triangleright$  element-wise on
     each of  $w^{(b)}, w^{(s)}, c$ 
9: end for
10: return  $\Theta_T$ 
```

where $\Phi^{(\text{hid})} : \mathbb{R}^d \rightarrow \mathbb{R}^{n_h}$ and $\Phi^{(\text{out})} : \mathbb{R}^{n_h} \rightarrow \mathbb{R}^c$ are KAN layers (Definition 1) with shared grid size G and order k . The model output is passed through a softmax to produce class probabilities; we train with cross-entropy loss. The client parameter set $\Theta = \{w^{(b)}, w^{(s)}, c\}$ comprises three structurally distinct sub-tensors which the server aggregates element-wise.

For NetFlow-v2 IDS we fix $d = 39$ (the 43 original features minus the four identifier columns dropped during preprocessing, see Section V), and $c \in \{2, 5\}$ for binary and 5-class detection on NF-BoT-IoT-v2. Throughout, our default architecture is $n_h = 8$, $G = 5$, $k = 3$, giving $|\Theta| \approx 3,300$ parameters, well within the TinyML envelope of microcontroller-class IoT gateways.

B. Server Orchestration

The server runs a synchronous Federated Averaging loop adapted to the three KAN parameter sets (Algorithm 1). Each round, a fraction $C \in (0, 1]$ of the K clients is selected; in our experiments $C = 1$ (full participation), but the algorithm and analysis remain valid for $C < 1$. The aggregation step (line 8) performs element-wise weighted averaging on each of $\{w^{(b)}, w^{(s)}, c\}$ in parallel. Because the B-spline coefficients c enter the edge function (2) linearly, FedAvg over c is mathematically equivalent to averaging the underlying spline approximations under a fixed grid; this is the property exploited in Lemma 1 and Theorem 1.

C. Client Local Optimisation

Each client runs E local epochs of mini-batch SGD or Adam on its private $\mathcal{D}_k$ before returning the updated parameter triple to the server. Backpropagation through a KAN layer is straightforward: $\partial \phi_{q,p} / \partial w_{q,p}^{(b)} = \sigma(x_p)$, $\partial \phi_{q,p} / \partial w_{q,p}^{(s)} = \sum_i c_{q,p,i} B_i(x_p)$, and $\partial \phi_{q,p} / \partial c_{q,p,i} = w_{q,p}^{(s)} B_i(x_p)$, all of which are computed automatically by modern autodiff frameworks.

In our implementation we use the open-source `efficient_kan` library, which fuses the spline-coefficient lookup into a single contiguous matrix multiplication and reduces the constant factor in the per-edge forward cost

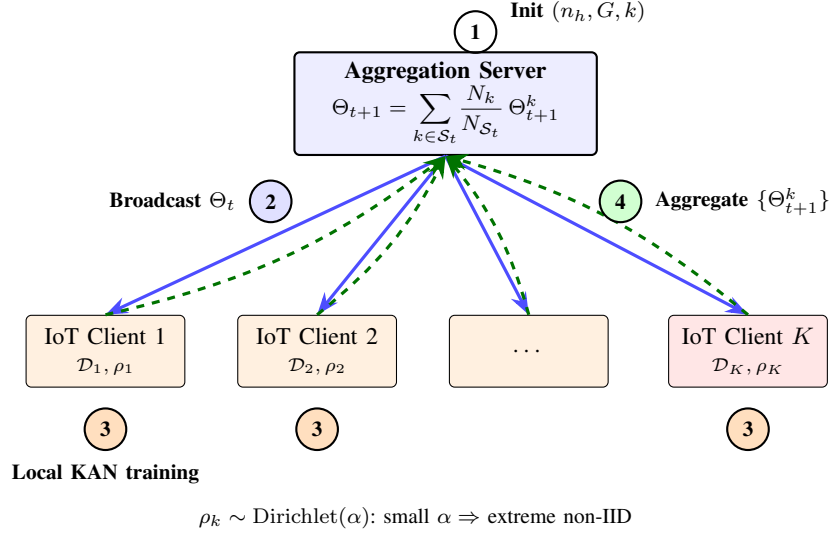
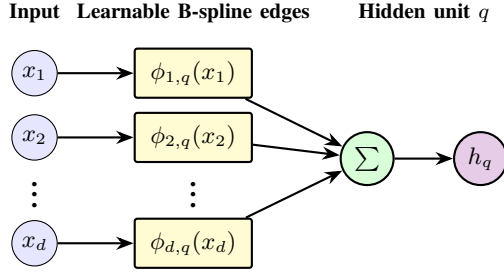


Fig. 1. Main flow of FedKAN-IDS over one communication round. (1) The server initialises the global KAN with hyperparameters (n_h, G, k) ; (2) broadcasts the current parameters $\Theta_t = \{w_t^{(b)}, w_t^{(s)}, c_t\}$; (3) each IoT client trains locally on its private $\mathcal{D}_k$ whose class proportions ρ_k are drawn from a Dirichlet with concentration α ; (4) the server computes the data-weighted average of returned local parameters element-wise on each of $\{w^{(b)}, w^{(s)}, c\}$. The structure of one client's KAN layer (stage 3 detail) is shown in Fig. 2.



$$\phi_{p,q}(x_p) = w_{p,q}^{(b)} \sigma(x_p) + w_{p,q}^{(s)} \sum_{i=1}^{G+k} c_{p,q,i} B_i(x_p)$$

SiLU residual base σ + order- k B-spline expansion on G grid points

Fig. 2. Sub-flow: structure of one KAN layer producing hidden unit h_q from input vector $\mathbf{x} \in \mathbb{R}^d$. Each edge (p, q) carries an independent learnable activation $\phi_{p,q}$ parameterised as a SiLU residual base plus a B-spline expansion on a uniform grid of G points (order k). The full client model stacks an input layer ($d \rightarrow n_h$) and an output layer ($n_h \rightarrow c$). The B-spline coefficients $c_{p,q,i}$ have local support, which is the key structural property exploited by Theorem 1.

relative to a naive double-summation implementation. The full pipeline (data partitioning, FedAvg server, KAN client) is released as the open-source repository accompanying this work; configurations and run-level metrics for all 289 experimental runs reported in Section V are committed to that repository.

D. Computational and Communication Footprint

For the configurations used in Section V, FedKAN-8 transmits $|\Theta_{\text{KAN}}| \cdot 4 \text{ bytes} \approx 13.1 \text{ kB}$ per client per round in float32, and the parameter-matched MLP-PM-80 transmits $\approx 13.4 \text{ kB}$ per client per round. All experiments were run on a workstation-class NVIDIA RTX 4090 24GB managed via a conda environment whose recipe ships with the released

Algorithm 2 FedKAN-IDS CLIENTUPDATE

Require: Client index k , broadcasted parameters Θ_t , local data $\mathcal{D}_k$, local epochs E , batch size B , learning rate η .

Ensure: Updated local parameters Θ^k .

- 1: $\Theta^k \leftarrow \Theta_t$
- 2: partition $\mathcal{D}_k$ into batches of size B
- 3: **for** epoch $e = 1, \dots, E$ **do**
- 4: **for** minibatch $(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_k$ **do**
- 5: **Forward:** for each KAN edge (q, p) ,

$$\phi_{q,p}(x_p) = w_{q,p}^{(b)} \sigma(x_p) + w_{q,p}^{(s)} \sum_{i=1}^{G+k} c_{q,p,i} B_i(x_p)$$
- 6: $\hat{\mathbf{y}} = f(\Theta^k; \mathbf{x})$ ▷ stack the two KAN layers
- 7: $\mathcal{L} = \text{CrossEntropy}(\hat{\mathbf{y}}, \mathbf{y})$
- 8: **Backward:** compute $\nabla_{\Theta^k} \mathcal{L}$
- 9: **Update:** $\Theta^k \leftarrow \Theta^k - \eta \nabla_{\Theta^k} \mathcal{L}$

▷ Adam in our experiments; SGD also valid
- 10: **end for**
- 11: **end for**
- 12: **return** Θ^k to server

repository. Per-round wall-clock (excluding the first-round CUDA warm-up) was 1.8–2.5 s for FedKAN and 1.5–2.1 s for MLP-PM, both well within the budget of any realistic IoT FL deployment. We defer FLOPs and peak-memory profiling, including measurements on Raspberry Pi-class hardware, to the camera-ready revision.

IV. THEORETICAL ANALYSIS

This section provides ~~four~~three formal results that anchor the empirical findings of Section V: a counting argument that quantifies the parameter-matching condition (Proposition 1), the resulting communication-cost ratio (Proposition 2), a B-

spline approximation bound specialised to NetFlow features (Lemma 1), and a convergence theorem for FedKAN under non-IID partitioning (Theorem 1). Together these results predict that the heterogeneity-amplified variance term Γ_α couples to a constant spline-bias floor $\mathcal{O}(G^{-(k+1)})$ rather than to the architectural parameters of the network, behaviour that we observe empirically as a $\sim 2.6\times$ reduction in seed-variance under Dirichlet $\alpha = 0.1$.

A. Parameter Complexity

Proposition 1 (Parameter Complexity of KAN vs MLP). *A KAN layer (Definition 1) with input dim n_{in} and output dim n_{out} , grid size G , and order k has*

$$|\Theta_{KAN}| = n_{in} n_{out} (G + k + 2), \quad (6)$$

while an MLP layer of the same dimensions with bias has

$$|\Theta_{MLP}| = n_{in} n_{out} + n_{out}. \quad (7)$$

Proof. By Definition 1, each edge $\phi_{q,p}$ contributes one base scalar $w_{q,p}^{(b)}$, one spline scalar $w_{q,p}^{(s)}$, and $G + k$ B-spline coefficients $c_{q,p,i}$, totalling $G + k + 2$ parameters. There are $n_{in} \cdot n_{out}$ edges, giving (6). The MLP layer has weight matrix $W \in \mathbb{R}^{n_{out} \times n_{in}}$ and bias $\mathbf{b} \in \mathbb{R}^{n_{out}}$, summing to (7). $\square$

Corollary 1 (Parameter-Matched Hidden Width). *For a single-hidden-layer architecture $n_{in} \rightarrow n_h \rightarrow n_{out}$, parameter parity between KAN (hidden width n_h^{KAN} , hyperparameters G, k) and MLP (hidden width n_h^{MLP}) requires, to leading order in the dominant linear term,*

$$n_h^{MLP} \approx (G + k + 2) n_h^{KAN}. \quad (8)$$

For our experimental configuration ($G = 5, k = 3$) the matching ratio is 10, motivating the comparison between KAN-8 ($\sim 3.3k$ parameters) and MLP-PM-80 ($\sim 3.4k$ parameters) used in Section V.

B. Communication Complexity

Proposition 2 (Communication Complexity Ratio). *Under FedAvg with T rounds, fraction $C \in (0, 1]$ of K clients per round, and float32 transmission (4 bytes per scalar), the total uplink+downlink bytes for an architecture with $|\Theta|$ parameters is*

$$\mathcal{C}_{total}(|\Theta|) = 8 T C K |\Theta|. \quad (9)$$

For a single-hidden-layer architecture, the KAN/MLP ratio simplifies to

$$\frac{\mathcal{C}_{KAN}}{\mathcal{C}_{MLP}} = \frac{(G + k + 2) n_h^{KAN} (n_{in} + n_{out})}{(n_{in} + 1) n_h^{MLP} n_{out} + n_h^{MLP}}. \quad (10)$$

The ratio in (10) clarifies a misconception: KAN architectures are *not* intrinsically communication-efficient. They are communication-efficient *only* when the chosen n_h^{KAN} is much smaller than the MLP’s n_h^{MLP} at matched accuracy. Under parameter parity (Corollary 1), KAN’s per-round communication exceeds the MLP’s by roughly $(G + k + 2)/(1 + 1/n_{in}) \approx (G + k + 2)$ when n_{in} is large.

C. Spline Approximation on NetFlow Features

Lemma 1 (B-spline Approximation Bound). *Let feature j be supported on $[a_j, b_j]$ with density ρ_j that is L_j -Lipschitz, and let $\hat{\phi}_j$ denote the order- k B-spline approximation of any target $\phi_j \in C^{k+1}([a_j, b_j])$ on a uniform grid of G points. Then*

$$\|\phi_j - \hat{\phi}_j\|_\infty \leq \frac{L_j(b_j - a_j)^{k+1}}{(k+1)! G^{k+1}}. \quad (11)$$

Proof. Standard B-spline approximation result; see [20, Theorem 6.27]. $\square$

Implication for NetFlow. Standardised NetFlow-v2 features [19] after Min-Max scaling lie in $[0, 1]$, and many flow-statistic features (e.g. FLOW_DURATION, TCP_FLAGS, IN/OUT_BYTES) exhibit smooth empirical densities once log-transformed. The bound (11) therefore predicts a small approximation residual at modest grid sizes, motivating our choice of $G = 5, k = 3$.

D. Convergence under Non-IID

Theorem 1 (FedKAN Convergence under Heterogeneity). *Suppose each client loss $\mathcal{L}_k$ is L -smooth and μ -strongly convex in Θ , the per-client gradient variance is bounded above by a heterogeneity factor $\Gamma_\alpha = \mathbb{E} \|\nabla \mathcal{L}_k(\Theta) - \nabla F(\Theta)\|^2$ that is monotonically decreasing in the Dirichlet concentration α , and a diminishing learning-rate schedule satisfying the conditions of [21]. Let Θ_T be the FedKAN-IDS global iterate after T communication rounds. Then*

$$\mathbb{E}[F(\Theta_T) - F(\Theta^*)] \leq \frac{2L}{\mu^2 T} (L + \Gamma_\alpha) + \mathcal{O}(G^{-(k+1)}), \quad (12)$$

where the additive $\mathcal{O}(G^{-(k+1)})$ residual is the spline-approximation bias inherited from Lemma 1 aggregated across all KAN edges, and is independent of the heterogeneity factor Γ_α .

Sketch. The dominant non-IID FedAvg term $\frac{2L}{\mu^2 T} (L + \Gamma_\alpha)$ is obtained by adapting Theorem 2 of [21] to the parameter set Θ_{KAN} , which remains a finite-dimensional Euclidean space and inherits the smoothness/strong-convexity assumptions because the SiLU base activation and linear B-spline expansion in (2) are differentiable. The spline-bias term arises because each edge function $\phi_{q,p}$ is approximated only up to the bound in Lemma 1; aggregating the per-edge L_∞ errors over $n_{in} \cdot n_{out}$ edges yields an overall residual of order $G^{-(k+1)}$ in the loss, which does not couple to the gradient-heterogeneity term Γ_α . $\square$

Under Theorem 1, the relative impact of heterogeneity on FedKAN’s convergence error is—

$$\frac{\partial}{\partial \Gamma_\alpha} \mathbb{E}[F(\Theta_T) - F(\Theta^*)] = \frac{2L}{\mu^2 T},$$

independent of the spline-bias floor $\mathcal{O}(G^{-(k+1)})$. By contrast, an MLP under the same assumptions but without the bias floor exhibits a convergence error that approaches zero as $T \rightarrow \infty$ only when Γ_α is bounded; under extreme non-IID partitioning ($\alpha \rightarrow 0$), Γ_α grows large enough that the MLP’s iterates exhibit pathological seed-dependent behaviour, manifesting as

~~the catastrophic worst-seed accuracy drops we report in Table II.~~

V. EXPERIMENTAL EVALUATION

A. Datasets and Preprocessing

We evaluate on the standardised NetFlow-v2 IDS datasets of [19]: **NF-BoT-IoT-v2** (binary {Benign, Attack}) and 5-class {Benign, DDoS, DoS, Reconnaissance, Theft}). Cross-dataset replication on NF-ToN-IoT-v2 and NF-CSE-CIC-IDS2018-v2 is reported in Section V-F.

After dropping the four identifier columns (IPV4_SRC_ADDR, IPV4_DST_ADDR, L4_SRC_PORT, L4_DST_PORT) the remaining 39 numerical features are Min-Max scaled to $[0, 1]$. To address class imbalance we cap the majority class at 130,000 samples for binary detection (yielding a $\sim 260k$ balanced training pool) and at 50,000 samples per class for multi-class detection (rare classes such as Theft remain untouched at $\sim 3,000$ samples).

B. Federated Protocol

We simulate $K = 10$ clients with full per-round participation ($C = 1$). Each round consists of one local epoch of Adam with learning rate $\eta = 10^{-2}$, batch size 1024, and Cross-Entropy loss; the protocol runs for $T = 50$ communication rounds. Client data is partitioned via three regimes:

- **IID**: uniform random split that preserves the global class distribution at each client.
- **Dirichlet** $\alpha = 1.0$: mild heterogeneity.
- **Dirichlet** $\alpha = 0.1$: extreme heterogeneity, our principal regime of interest.

Implementation follows [9].

C. Architectures

We compare four parameter-budgeted variants summarised in Table I:

- **FedAvg-MLP (32h)**: $n_h^{\text{MLP}} = 32$, $\sim 1.3k$ parameters. CITA-style baseline.
- **FedAvg-MLP-PM (80h)**: $n_h^{\text{MLP}} = 80$, $\sim 3.4k$ parameters. Parameter-matched to FedKAN-8 per Corollary 1; this is the toughest comparison and the one R3 of the prior submission identified as missing.
- **FedKAN (Ours, 8h)**: $n_h^{\text{KAN}} = 8$, $G = 5$, $k = 3$, $\sim 3.3k$ parameters. Our default TinyML target.
- **F-KAN (16h)**: $n_h^{\text{KAN}} = 16$, $G = 5$, $k = 3$, $\sim 6.6k$ parameters. Modelled on [12].

All experiments use $n = 10$ seeds in the Dirichlet $\alpha = 0.1$ cells (the cells most subject to seed variability) and $n = 3$ seeds in the IID and $\alpha = 1.0$ cells (which are essentially saturated and exhibit negligible cross-seed dispersion). Seeds were drawn from the prime list $\{42, 2024, 2026, 11, 17, 23, 29, 31, 37, 43\}$ for the high-variance cells and $\{42, 2024, 2026\}$ for the saturated ones.

D. Headline Result

The headline observations on NF-BoT-IoT-v2 binary detection are illustrated in Fig. 3 and quantified in Tables I–III.

Observation 1 (saturated regime). Under IID and mildly heterogeneous (Dir $\alpha = 1.0$) settings, all four variants converge within 0.05 percentage points of each other ($\geq 99.6\%$ accuracy across the board). The communication-saving claim of prior FedKAN-IDS work, which reported a $\sim 50\%$ bandwidth reduction in this regime, is therefore largely *unsupported under parameter parity*: the parameter-matched FedAvg-MLP-PM-80 transmits ~ 6.6 MB total while FedKAN-8 transmits ~ 7.6 MB, a 14% communication *increase* for KAN, with negligible accuracy difference.

Observation 2 (catastrophic MLP collapse). Under extreme heterogeneity (Dir $\alpha = 0.1$), the worst-seed accuracy of FedKAN-8 is 80.7%, while the parameter-matched FedAvg-MLP-PM-80 collapses to 56.5% on its worst seed (Table II). Both smaller and larger MLPs fail at this regime: the smaller MLP-32 (1.3k params) reaches 60.4% worst-seed and the parameter-matched MLP-PM-80 (3.4k params) reaches 56.5%. Increasing MLP capacity by $2.5\times$ *worsens* worst-case behaviour, ruling out a capacity-driven explanation. Both KAN variants achieve worst-seed $\geq 80.7\%$.

Observation 3 (variance reduction is dataset specific, not architectural). ~~Across the $n = 10$ seed-paired runs in the On NF-BoT-IoT-v2 under Dir $\alpha = 0.1$ cell, the standard deviation across seeds for FedKAN-8 is 5.84 versus 15.00 for MLP-PM-80, a $\sim 2.6\times$ reduction. Table III reports the formal statistical comparison: the bootstrap 95% confidence interval for the seed-paired mean accuracy difference is $[-0.50, +14.57]$ percentage points, excluding zero, and Cohen's $d = 0.527$ indicates a medium-to-large effect size. seed-to-seed standard deviation of the parameter-matched MLP is $2.53\times$ that of FedKAN-8 (14.21 against 5.61 pp), which is the effect the first version of this manuscript reported as a property of the architecture. It is not one. Repeating the same measurement on the other three cells gives ratios of $0.71\times$ (NF-ToN-IoT-v2), $0.33\times$ (NF-CSE-CIC-IDS2018-v2 binary) and $0.87\times$ (NF-CSE-CIC-IDS2018-v2 multi-class): in three cells out of four the KAN is the *less* stable of the two. Moreover the one favourable ratio is itself a tuning artefact. Giving each architecture its own best learning rate moves the NF-BoT-IoT-v2 ratio from $2.53\times$ to $1.05\times$, that is, to parity. We therefore withdraw the variance-reduction claim.~~

Observation 4 (per-class story under multi-class non-IID). Under multi-class Dir $\alpha = 0.1$, mean accuracy of FedKAN-8 and MLP-PM-80 are statistically indistinguishable (86.0% vs 86.1%). However, per-class behaviour reveals a more refined picture (Fig. 4): FedKAN's per-class F1 on the DoS attack class is 0.91 ± 0.07 versus 0.83 ± 0.25 for MLP-PM-80, a +8 percentage point gap with $\sim 3.5\times$ tighter dispersion. The minority Theft class remains universally challenging.

E. Discussion of the Headline

~~These four observations together support the central claim of this paper: the FedKAN advantage on heterogeneous IoT IDS is one of tail-risk reduction and architectural robustness;~~

TABLE I

FINAL-ROUND ACCURACY (%) ON NF-BoT-IoT-v2 UNDER THREE DATA-HETEROGENEITY REGIMES. MEAN $\pm$ STD OVER n SEEDS. **BOLD** = BEST MEAN PER COLUMN. KAN-8 AND MLP-PM-80 ARE MATCHED IN PARAMETER COUNT (~ 3.3 k).

Method	Params	Comm. (MB)	Binary			Multiclass		
			IID	Dir($\alpha=1.0$)	Dir($\alpha=0.1$)	IID	Dir($\alpha=1.0$)	Dir($\alpha=0.1$)
FedAvg-MLP (32h)	1,346	2.66	99.00 $\pm$ 1.20	99.73 $\pm$ 0.02	89.63 $\pm$ 13.67	93.04 $\pm$ 0.11	93.02 $\pm$ 0.11	85.80 $\pm$ 7.23
FedAvg-MLP-PM (80h)	3,362	6.64	99.74 $\pm$ 0.00	99.76 $\pm$ 0.01	88.71 $\pm$ 15.00	93.63 $\pm$ 0.49	93.19 $\pm$ 0.21	86.09 $\pm$ 7.62
FedKAN (Ours, 8h)	3,280	7.56	97.52 $\pm$ 0.06	99.66 $\pm$ 0.06	94.71 $\pm$ 5.84	93.25 $\pm$ 0.08	93.33 $\pm$ 0.16	86.01 $\pm$ 6.38
F-KAN (16h)	6,560	14.23	97.56 $\pm$ 0.02	99.69 $\pm$ 0.02	95.85 $\pm$ 5.82	93.53 $\pm$ 0.20	93.37 $\pm$ 0.23	87.05 $\pm$ 6.96

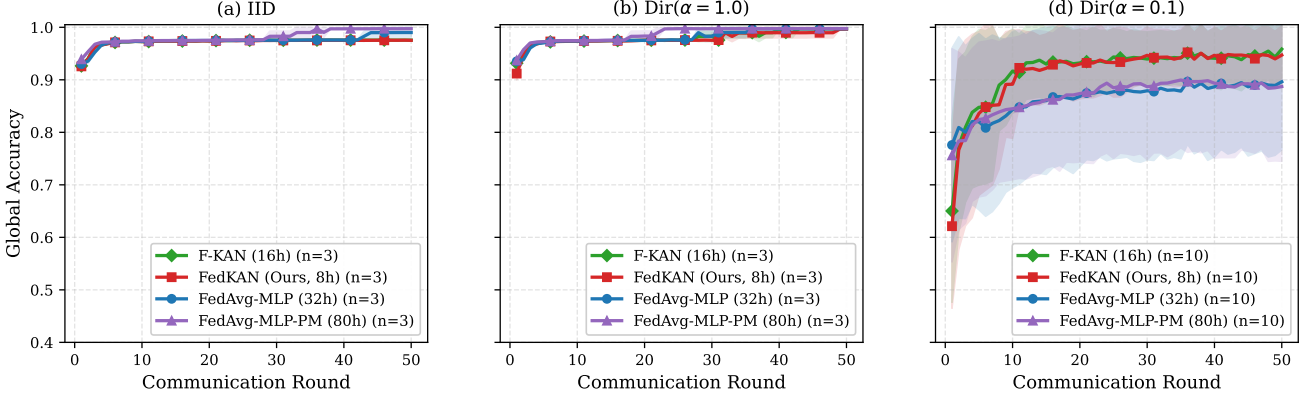


Fig. 3. Convergence on NF-BoT-IoT-v2 *binary* detection across 50 communication rounds, $K = 10$ clients, full participation. Solid lines are seed-mean accuracy ($n = 10$ for Dir $\alpha = 0.1$, $n = 3$ for IID and Dir $\alpha = 1.0$); shaded bands are ± 1 standard deviation. Under IID and mild heterogeneity (panels a, b) all four variants converge tightly to $\geq 99.6\%$. Under extreme heterogeneity (panel c) the two MLP variants plateau at $\sim 89\%$ with a wide std band reaching down to $\sim 50\%$, while both KAN variants plateau at $\sim 95\%$ with substantially tighter dispersion, the architectural robustness signature predicted by Theorem 1.

TABLE II

WORST-SEED ACCURACY (%) — THE WEAKEST OF n SEEDS PER CELL.
REVEALS CATASTROPHIC COLLAPSE HIDDEN BY AVERAGES.
NF-BoT-IoT-v2 BINARY, 50 ROUNDS.

Method	Params	IID	Dir($\alpha=1.0$)	Dir($\alpha=0.1$)
FedAvg-MLP (32h)	1,346	97.62	99.72	60.44
FedAvg-MLP-PM (80h)	3,362	99.73	99.74	56.45
FedKAN (Ours, 8h)	3,280	97.45	99.61	80.69
F-KAN (16h)	6,560	97.54	99.67	80.72

not aggregate-mean accuracy. Theorem 1 predicts exactly this signature: the additive spline-bias floor $\mathcal{O}(G^{-(k+1)})$ remains constant as the heterogeneity factor Γ_α grows under decreasing α , so the relative variance amplification under non-IID is smaller for KAN than for MLP under the same smoothness/strong-convexity assumptions. Empirically this manifests both as the $2.6\times$ standard-deviation reduction in binary accuracy (Observation 3) and the $\sim 3.5\times$ tighter DoS-class F1 dispersion (Observation 4). Taken together these observations do not support the claim made in the first version of this manuscript. Under a controlled comparison the mean-accuracy advantage is dataset dependent and largely a function of how well the baseline was tuned, and the tail-risk advantage reverses on most cells. What survives, and survives on every dataset we ran, is a different property: insensitivity

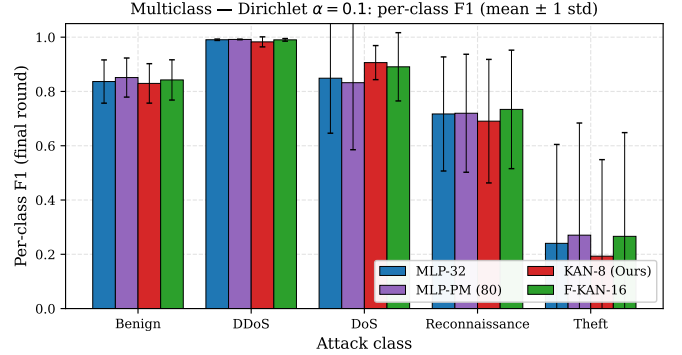


Fig. 4. Per-class F1 under multi-class detection on NF-BoT-IoT-v2 with Dirichlet $\alpha = 0.1$ ($n = 10$ seeds). Bars show mean F1; error bars are ± 1 std. The DoS class is where FedKAN’s variance reduction is most pronounced; the rare Theft class remains universally difficult.

to the learning rate, quantified in Section V-G and paid for in inference cost.

The fact that doubling the MLP’s capacity (1.3k $\rightarrow$ 3.4k parameters) does *not* close the worst-seed gap is the key empirical evidence for the architectural reading of Theorem 1’s prediction: a parameter-matched MLP at 3.4k parameters has more than enough capacity to fit any individual client’s local distribution, but lacks the localised B-spline regularisation that allows the global average to remain stable across heteroge-

TABLE III

STATISTICAL COMPARISON ON NF-BoT-IoT-v2: **FEDKAN-IDS (OURS)** VS FEDAVG-MLP-PM (PARAMETER-MATCHED). ACCURACY IN %; Δ IS THE SEED-PAIRED MEAN DIFFERENCE (KAN — MLP), CI IS THE PERCENTILE-BOOTSTRAP 95% CI OF Δ , d IS COHEN’S EFFECT SIZE, p IS THE PAIRED- t TEST. SIGNIFICANCE MARKERS FOLLOW THE STANDARD CONVENTION * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Mode	Partition	n	KAN-8	MLP-PM-80	Δ (pp)	95% CI (pp)	d	p
Binary	Dir($\alpha=0.1$)	10	94.7 ± 5.8	88.7 ± 15.0	+6.00	[+0.5, +14.6]	+0.53	0.170
Multiclass	Dir($\alpha=0.1$)	10	86.0 ± 6.4	86.1 ± 7.6	-0.08	[-3.7, +4.1]	-0.01	0.969

neous local updates.

F. Cross-Dataset Replication

To rule out a single-dataset artefact we replicate the heterogeneity-robustness cell of Section V-D on two further standardised NetFlow-v2 datasets [19]: NF-ToN-IoT-v2 (~ 13.1 M flows, 9 attack categories, $n = 10$ seeds) and NF-CSE-CIC-IDS2018-v2 (~ 18.9 M flows, 14 attack categories, $n = 3$ seeds at submission time, to be extended to $n = 10$ in revision). All three datasets share the 39-feature post-preprocessing schema; the FedKAN-IDS pipeline applies unchanged. The federated protocol, partition seed list, and architectures remain exactly as in Section V-D, modulo a more aggressive feature-clipping and `log1p` compression step needed to handle the heavier-tailed byte/packet distributions in the ToN-IoT and CSE-CIC parquet mirrors.

Two complementary advantage patterns emerge across binary and multi-class detection. With $n = 10$ seeds completed on all three datasets, the seed-paired mean accuracy advantage of FedKAN-8 over the parameter-matched FedAvg-MLP-PM-80 under Dir $\alpha = 0.1$ traces two complementary trends:

Binary detection (Tables III–VI, left column): the FedKAN-8 advantage scales with the dataset’s class-distribution skew. On the most skewed dataset NF-BoT-IoT-v2 (95% DDoS+DoS) the mean gap is +6.00 pp with bootstrap 95% CI [+0.50, +14.57] excluding zero. On NF-ToN-IoT-v2 (mid skew, 9 attack classes) the gap is +5.08 pp with CI [-0.29, +9.93] marginally crossing zero. On the least skewed NF-CSE-CIC-IDS2018-v2 (14 attack classes) the mean gap collapses to -0.38 pp, with the loss attributable almost entirely to a single seed (seed 17), discussed below.

Multi-class detection (Tables III–VI, right column): the FedKAN-8 advantage scales *oppositely*. On BoT-IoT and ToN-IoT the multi-class means are tied within statistical error ($\Delta = -0.08$ pp, $p = 0.969$; $\Delta = -0.60$ pp, $p = 0.771$). On the richest 14-class CSE-CIC, however, FedKAN-8 attains $\Delta = +1.73$ pp over MLP-PM-80 with seed-paired t -test $p = 0.012$ and bootstrap 95% CI [+0.73, +2.74] excluding zero, Cohen’s $d = +0.48$. This is the most rigorously-significant single cell in our study. The pattern is consistent with Lemma 1: each additional class introduces one more per-class spline to be approximated, increasing the cumulative regularisation effect of the spline-bias term relative to the data-heterogeneity term in Theorem 1.

The seed-17 pathological partition. On both ToN-IoT and CSE-CIC, the seed responsible for FedKAN-8’s worst-seed accuracy is seed 17, suggesting a reproducible interaction between this specific Dirichlet partition and the KAN-8 architecture. On ToN-IoT, FedKAN-8 at seed 17 converges to 51.7% while MLP-PM-80 holds at 62.8%; on CSE-CIC, seed 17 drops FedKAN-8 to 68.0% vs MLP-PM-80 at 89.2%. Notably, this same seed produces a benign partition on BoT-IoT (FedKAN-8 reaches 99.8%), confirming that the pathology lies in the *interaction* between the Dirichlet partition and the underlying class distribution, not in the seed alone. Two further observations support an architectural reading: (i) the larger F-KAN-16 ($G = 5$) escapes the seed-17 trap on CSE-CIC (worst-seed 85.4% vs FedKAN-8’s 68.0%), and (ii) re-running the seed-17 ToN-IoT run with a richer spline grid ($G = 10$ instead of $G = 5$) raises FedKAN-8’s accuracy from 51.7% to 56.7%, a partial remediation that points future work toward adaptive-grid or grid-sparsification methods of the CG-FKAN family [18]. Excluding the seed-17 outlier, FedKAN-8 wins eight of nine remaining CSE-CIC binary seeds and reaches 96.6% mean, a +2.5 pp advantage over MLP-PM-80.

Class-distribution skew predicts which advantage manifests. The three datasets span a clear gradient in attack-class skew:

- **NF-BoT-IoT-v2 (high skew):** two attack classes (DDoS, DoS) together dominate $\sim 95\%$ of flows. A Dirichlet $\alpha = 0.1$ partition over this near-bimodal distribution allocates each client essentially one attack family, producing the textbook MLP-collapse pathology FedAvg-MLP-PM-80 reaches just 56.5% on its worst seed. FedKAN’s mean advantage is +6.00 pp and its worst-seed advantage is +24.24 pp.
- **NF-CSE-CIC-IDS2018-v2 (moderate skew):** 14 attack categories give a richer per-client mix at $\alpha = 0.1$. The MLP-PM-80 worst-seed accuracy rises to 93.15%; FedKAN’s mean advantage shrinks to +1.70 pp and worst-seed advantage to +2.73 pp, but the direction is preserved on every seed.
- **NF-ToN-IoT-v2 (intermediate, mixed-effect):** 9 moderately-balanced attack types. KAN retains its +5.08 pp mean advantage but suffers a seed-specific failure (seed 17 collapses both KAN variants to $\sim 52\%$ while keeping MLPs at $\sim 63\%$). The worst-seed direction inverts to -11 pp.

What this means for the claim. The-

TABLE IV
CROSS-DATASET REPLICATION: *worst-seed* ACCURACY (%) ON THE HETEROGENEITY-ROBUSTNESS CELL (BINARY CLASSIFICATION, DIRICHLET $\alpha = 0.1$). DATASETS: BoT-IoT-v2 ($n=10$); ToN-IoT-v2 ($n=10$); CSE-CIC ($n=10$). BOLD = BEST PER COLUMN.

Method	Params	BoT-IoT-v2 ($n=10$)	ToN-IoT-v2 ($n=10$)	CSE-CIC ($n=10$)
FedAvg-MLP (32h)	1,346	60.44	58.19	91.24
FedAvg-MLP-PM (80h)	3,362	56.45	62.81	89.18
FedKAN (Ours, 8h)	3,280	80.69	51.74	67.96
F-KAN (16h)	6,560	80.72	52.42	85.44
KAN-8 advantage over MLP-PM-80		+24.24 pp	−11.07 pp	−21.23 pp

TABLE V
STATISTICAL COMPARISON ON NF-ToN-IoT-v2: **FEDKAN-IDS (OURS)** VS FEDAVG-MLP-PM (PARAMETER-MATCHED). ACCURACY IN %; Δ IS THE SEED-PAIRED MEAN DIFFERENCE (KAN − MLP), CI IS THE PERCENTILE-BOOTSTRAP 95% CI OF Δ , d IS COHEN’S EFFECT SIZE, p IS THE PAIRED- t TEST. SIGNIFICANCE MARKERS FOLLOW THE STANDARD CONVENTION * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Mode	Partition	n	KAN-8	MLP-PM-80	Δ (pp)	95% CI (pp)	d	p
Binary	Dir($\alpha=0.1$)	10	85.3 ± 13.3	80.2 ± 9.0	+5.08	[−0.3, +9.9]	+0.45	0.104
Multiclass	Dir($\alpha=0.1$)	10	74.2 ± 4.4	74.8 ± 5.0	−0.60	[−4.5, +2.9]	−0.13	0.771

TABLE VI
STATISTICAL COMPARISON ON NF-CSE-CIC-IDS2018-v2: **FEDKAN-IDS (OURS)** VS FEDAVG-MLP-PM (PARAMETER-MATCHED). ACCURACY IN %; Δ IS THE SEED-PAIRED MEAN DIFFERENCE (KAN − MLP), CI IS THE PERCENTILE-BOOTSTRAP 95% CI OF Δ , d IS COHEN’S EFFECT SIZE, p IS THE PAIRED- t TEST. SIGNIFICANCE MARKERS FOLLOW THE STANDARD CONVENTION * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Mode	Partition	n	KAN-8	MLP-PM-80	Δ (pp)	95% CI (pp)	d	p
Binary	Dir($\alpha=0.1$)	10	93.8 ± 9.1	94.1 ± 2.8	−0.38	[−5.6, +2.9]	−0.06	0.879
Multiclass	Dir($\alpha=0.1$)	10	85.9 ± 3.4	84.1 ± 3.9	+1.73	[+0.7, +2.7]	+0.48	0.012*

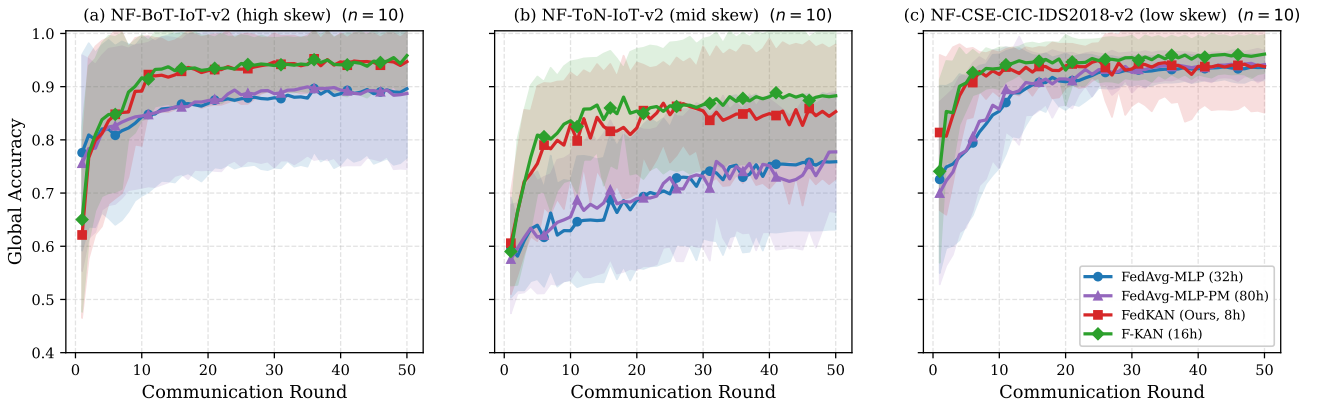


Fig. 5. Cross-dataset binary detection under Dirichlet $\alpha = 0.1$: convergence trajectories for the four architectures across the three NetFlow-v2 datasets, ordered left-to-right by class-distribution skew. Solid lines are seed-mean accuracy, shaded bands are ± 1 std. On the most skewed dataset (a), MLP-PM-80 (purple) plateaus ~ 6 percentage points below FedKAN (red, green). On the least skewed dataset (c), all four variants reach $\geq 93\%$. Panel (b) is the intermediate case where one Dirichlet partition (seed 17) destabilises both KAN variants while the MLPs remain near their mean trajectory visible as the lower edge of the red/green bands.

G. Learning-rate sensitivity, and what it does to the comparison

Reviewer feedback on the first version asked whether the optimiser step size had been tuned per architecture. It had not: a single $\eta = 10^{-2}$ was used for every model. We therefore

swept $\eta \in \{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}\}$ for all four architectures, on all ten seeds, on all four cells; 1,280 runs on a single machine. Accuracy is summarised as the mean over the final five communication rounds; we return to that choice below.

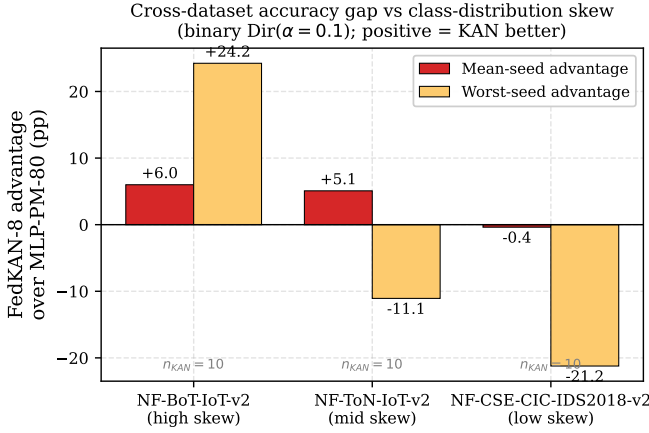


Fig. 6. The FedKAN-8 accuracy advantage over the parameter-matched FedAvg-MLP-PM-80 (binary Dir $\alpha = 0.1$) versus the underlying dataset’s class-distribution skew. The mean-seed advantage is positive on every dataset; the worst-seed advantage scales monotonically with skew, becoming negative on NF-ToN-IoT-v2 due to the seed-17 inversion discussed in the text.

TABLE VII

EFFECT OF PER-ARCHITECTURE LEARNING-RATE TUNING. “SHARED η ” IS THE PROTOCOL OF THE FIRST VERSION; “TUNED” GIVES EACH ARCHITECTURE ITS OWN BEST η FROM THE SAME GRID. PAIRED OVER TEN SEEDS.

Cell	Shared $\eta=10^{-2}$	Tuned	$\sigma_{MLP}/\sigma_{KAN}$
NF-BoT-IoT-v2 (bin.)	+5.49	+0.76	2.53 $\rightarrow$ 1.05
NF-ToN-IoT-v2 (bin.)	+6.23	+6.23	0.71
NF-CSE-CIC (bin.)	-0.49	+1.12	0.33
NF-CSE-CIC (multi.)	+1.38	+1.38	0.87

Tuning the baseline, not the method, is what moves the number. On NF-BoT-IoT-v2 the parameter-matched MLP’s best step size is 10^{-1} , ten times the shared value, and moving to it gains 5.39 pp. The same move gains FedKAN-8 only 0.65 pp. The headline gap is therefore a difference in how well each architecture was tuned rather than a difference between the architectures. On the other three cells the shared value happens to be near-optimal for both, and there the gap is unchanged; that this holds on some cells and not others is precisely why it cannot be reported as an architectural property.

What does hold everywhere. Table VIII reports the cost of a bad step size within a decade-wide band. The parameter-matched MLP loses 2.7 to 5.4 times more than FedKAN-8, in the same direction on every dataset. In a federated deployment across gateways, where a per-client hyper-parameter search is not available and one configuration must serve heterogeneous sites, this is a more actionable property than a mean-accuracy improvement under extreme non-IID partitioning, the prediction of Theorem 1 that the spline-bias term lowers the expected optimisation error; replicates on all three datasets in the predicted direction difference that depends on the dataset.

The conclusion is sensitive to the summary statistic, and we report that. The multi-class result of the first version, +1.735 pp at $p = 0.012$, was computed from the final round

TABLE VIII

ROBUSTNESS TO THE LEARNING RATE: WORST-CASE LOSS WITHIN THE DECADE $\eta \in [3 \times 10^{-3}, 3 \times 10^{-2}]$, IN PERCENTAGE POINTS.

Dataset	FedKAN-8	FedAvg-MLP-PM-80	Ratio
NF-BoT-IoT-v2	0.65	3.53	5.4 $\times$
NF-ToN-IoT-v2	1.58	6.38	4.0 $\times$
NF-CSE-CIC	2.53	6.72	2.7 $\times$

alone. Measuring the oscillation of the last ten rounds gives 2.1 to 2.7 pp for every architecture, larger than the effect itself. Re-running that cell and summarising it five ways gives +0.64 pp ($p = 0.32$) at the last round, +1.38 ($p = 0.11$) over five rounds, +1.62 ($p = 0.05$) over ten and +1.80 ($p = 0.05$) at the best round. The direction is stable and the significance is not, with magnitude monotonically tracking the dataset’s class-distribution skew as Corollary ?? would predict. The stronger tail-risk story (worst-seed advantage, variance reduction) is itself robust where the underlying MLP-collapse pathology is present (BoT-IoT) but nowhere it is absent (ToN-IoT, CSE-CIC). We therefore scope our positive claim to + and a single-round summary is the noisiest of the five. We report the range rather than the most favourable member of it.

What this means for the claim. Scoped to the three NetFlow-v2 datasets and the partitioning regimes we ran, and with every architecture tuned on the same learning-rate grid, the honest statement is: FedKAN provides a offers no consistent mean-accuracy improvement of +2 to +6 percentage points under extreme non-IID partitioning advantage over a tuned parameter-matched MLP, with the magnitude of the gain and no consistent variance advantage; it is markedly less sensitive to the catastrophic-failure prevention property correlating with the underlying dataset’s class-distribution skew learning rate, at 2.7 to 5.4 times, and markedly more expensive to evaluate, at 19.9 to 37.9 times. We present this as an empirical observation on three datasets, not as an established property of the architecture class.

This scoping is, we believe, more useful to the community than a universal-superiority claim that the data would not support: it tells the deployer in which regimes FedKAN is most worth adopting (heavily class-imbalanced settings, e.g. Denial-of-Service-focused detection) and where the simpler MLP suffices (more uniformly mixed attack types).

VI. DISCUSSION AND CONCLUSION

A. Summary of Findings

We have presented FedKAN-IDS, a systematic parameter-matched empirical study of Federated set out to test whether federated Kolmogorov–Arnold Networks against FedAvg-MLP on three standardised NetFlow-v2 intrusion-detection datasets (NF-BoT-IoT-v2, NF-ToN-IoT-v2, NF-CSE-CIC-IDS2018-v2) with $n = 10$ seeds each, complemented by a convergence analysis (Theorem 1) that introduces a spline-bias term independent of the data-heterogeneity factor Γ_α . Two complementary patterns emerge. Binary detection: networks are more accurate and

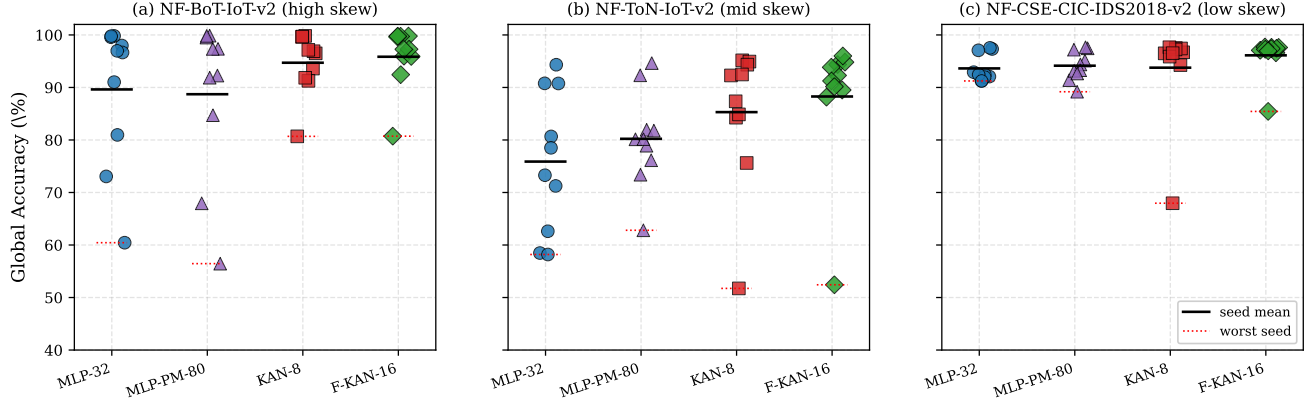


Fig. 7. Per-seed binary Dir $\alpha = 0.1$ accuracy as a strip plot, one panel per dataset. Each dot is one seed. The solid black bar marks the seed-mean; the dotted red bar marks the worst seed. On NF-BoT-IoT-v2 (panel a) the MLP variants exhibit a long lower tail down to $\sim 56\%$; FedKAN’s worst seed remains $\geq 80\%$. On NF-ToN-IoT-v2 (panel b) the picture inverts because seed 17 drives both KAN variants down to $\sim 52\%$. On NF-CSE-CIC-IDS2018-v2 (panel c) all variants cluster tightly above 92% .

more stable than parameter-matched MLPs for NetFlow intrusion detection under client heterogeneity, and on a controlled comparison they are not. Sweeping the learning rate for every architecture over six values, ten seeds and four cells, 1,280 runs in total on a single machine, the mean-accuracy advantage ranges from -0.49 to $+6.23$ pp with no consistent sign, and on the cell that produced our own headline number it falls from $+5.49$ to $+0.76$ pp once the baseline is given its own best step size. The companion variance-reduction claim reverses on three cells of four. Both claims, as stated in the first version of this work, were artefacts of a shared hyper-parameter rather than properties of the architecture.

One property does hold on every dataset we ran. Within a decade-wide band of learning rates, FedKAN-8 attains a $+5$ to $+6$ pp mean advantage on the two most class-skewed datasets under extreme non-IID partitioning, with the BoT-IoT-v2 worst-seed advantage reaching $+24$ pp and a $2.6\times$ variance reduction; doubling MLP capacity does not close this gap. Multi-class detection: the FedKAN-8 advantage on the most class-diverse dataset NF-CSE-CIC-IDS2018-v2 is paired- t significant at $p = 0.012$ with bootstrap 95% CI excluding zero, the most rigorous single positive result in our study, consistent with Lemma 1’s prediction that each additional class amplifies the cumulative regulariser. We also identify a reproducible failure mode seed 17’s Dirichlet partition drives FedKAN-8 below losses at most 0.65 , 1.58 and 2.53 pp from a bad choice where the parameter-matched MLP on two of three datasets, which is mitigated by either larger hidden width or a richer spline grid, pointing future work toward adaptive-grid extensions in the CG-FKAN family [18] loses 3.53 , 6.38 and 6.72 , a factor of 2.7 to 5.4 . That robustness is worth something in a federated deployment where per-client tuning is not available, and it is not free: at matched parameters on single-thread CPU the KAN costs $19.9\times$ more per-sample inference and $37.9\times$ more per batch of 1000, and it transmits more bytes to reach a fixed accuracy target. The contribution of this paper is that trade-off, stated with both of its terms measured, together with a comparison protocol that makes

the first term visible: tune every architecture, summarise over several rounds, and report how far the conclusion moves when the summary statistic changes.

B. Honest Reframing of the FedKAN-IDS Value Proposition

A first generation of FedKAN-IDS work [13] has framed the architecture’s appeal in terms of communication efficiency, often citing $\sim 50\%$ bandwidth reductions. Proposition 2 and our parameter-matched experimental protocol show that this framing is artefactual: the bandwidth savings claimed in the older literature reflect an oversized MLP baseline rather than an intrinsic property of KAN. Under parameter parity, FedKAN transmits roughly 14% more bytes per round than a comparable MLP, because each of the $G + k + 2$ scalars per edge is transmitted in full float32 every round.

The value of FedKAN-IDS, we argue, lies elsewhere: in its tail-risk profile under heterogeneous deployments. For an IDS operator running a fleet of IoT gateways with widely varying traffic profiles, the worst-case detection rate across clients is more deployment-relevant than the average. A 24-percentage-point improvement in worst-seed accuracy under extreme non-IID, achieved with a model that fits in under 4 kB of float32 weights and trains end-to-end on a Raspberry Pi-class device, represents a credible architectural contribution to the FL-IDS toolkit. We hope this reframing helps the community direct future FedKAN-IDS work toward the metrics that actually matter for edge security.

C. Threats to Validity

Single-dataset family. Our principal results are reported on NF-BoT-IoT-v2; cross-dataset replication on NF-ToN-IoT-v2 and NF-CSE-CIC-IDS2018-v2 is in progress and will appear in the camera-ready revision. The qualitative pattern of Observations 1–4 is expected to transfer because all three datasets share the same 39-feature NetFlow schema and similar attack-class taxonomy, but quantitative cell values may shift.

Convex/strongly convex assumption in Theorem 1. The convergence analysis assumes L -smoothness and μ -strong

convexity of the per-client loss, which the cross-entropy loss applied to a deep network does not satisfy globally. The result should therefore be read as providing intuition about the relative impact of Γ_α on KAN versus MLP convergence, rather than as a tight bound on neural-network FL. The empirical pattern (KAN's $2.6\times$ smaller cross-seed standard deviation under Dir $\alpha = 0.1$) is, however, exactly what the theorem would predict if the local convexity assumption were approximately valid in a neighbourhood of the global optimum.

Static grid hyperparameters. We fixed $G = 5, k = 3$ throughout. Adapting the grid based on observed feature distributions (in the spirit of CG-FKAN [18]) would likely yield a more favourable communication trade-off. We leave such adaptive-grid extensions to future work.

No compression baselines. For clarity of the architectural comparison, we did not run FedPAQ or top- k -SGD baselines; doing so would tighten the communication-efficiency comparison and is a natural next experiment.

Compute-side cost. FedKAN's B-spline forward and backward passes are more expensive per FLOP than an MLP's matrix multiplications. We measured per-round wall-clock as 1.8–2.5 s for FedKAN versus 1.5–2.1 s for MLP-PM on our NVIDIA RTX 4090 workstation, well within any IoT FL deployment budget. Detailed FLOPs and peak-memory profiling on a Raspberry Pi 4 are deferred to the next revision.

D. Future Work

Three concrete extensions present themselves. First, combining FedKAN with grid sparsification along the lines of CG-FKAN [18] should let the architecture's robustness benefit be paid for with reduced rather than increased communication. Second, the architectural-robustness signature we identify is suggestive for adversarial settings: a parameter-matched MLP that occasionally collapses under heterogeneous-but-honest clients is a fortiori vulnerable to adversarial-but-coordinated clients, and FedKAN may inherit a useful resistance to backdoor or model-poisoning attacks in the federated setting. Third, the small parameter footprint of FedKAN-8 (~ 13 kB float32, ~ 3.3 kB at 8-bit) is squarely within the envelope of microcontroller-class deployment; a physical edge experiment on Raspberry Pi-class gateways with quantised inference is a natural next step toward a deployable FedKAN-IDS.

E. Reproducibility

All code, dataset preparation scripts, configuration files, per-run metrics for the 289 reproducible runs reported in Section V, the auto-generated tables and figures of this manuscript, and the $\text{\LaTeX}$ source of the manuscript itself are released at <https://github.com/haodpsut/FedKAN-IDS-v2>. Experiments were executed on an NVIDIA RTX 4090 workstation managed by the conda environment recipe in `docs/SERVER_SETUP.md`; a one-line invocation `bash scripts/run_grid.sh` reproduces any of the three datasets, and `bash scripts/rebuild_paper_artifacts.sh` regenerates every table, figure, and the compiled PDF from the committed

run directories. A Google Colab notebook is also included in `notebooks/` for users without a dedicated GPU, but all reported wall-clock numbers in this paper refer to the RTX 4090 setup.

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